SOUND EVIDENCE · ONE-PAGE BRIEF

Music, sound & wellbeing *what the research supports*

A plain-language summary of 14 common claims, rated from 39 published sources. Educational only: not medical advice, and not a substitute for care from a qualified clinician.

Strong — Best supported

Anxiety & pain around surgery. Recorded music around surgery lowered anxiety and pain in reviews of 26, 73 and 92 randomized trials; one also found less pain-medication use. Hospital stays were not shorter. (Bradt et al. 2013; Hole et al. 2015; Kühlmann et al. 2018)

Moderate — Supported for specific, short-term uses

Everyday stress & anxiety. Across 104 randomized trials, music reduced how stressed people felt and short-term body measures such as heart rate. Effects were small to medium and mostly in the moment. (de Witte et al. 2020; de Witte et al. 2022)

Sleep. Daily music listening improved self-rated sleep quality in adults with insomnia symptoms (moderate certainty). Insomnia severity and objectively measured sleep did not clearly improve. (Jespersen et al. 2022)

Chronic & everyday pain. As an add-on, music lowered reported pain by about 1 point on a 0–10 scale across 97 trials. Self-chosen music did better; studies vary a lot. (Lee 2016; Garza-Villarreal et al. 2017; Chen et al. 2025)

Mood & depression. Music therapy added to usual care reduced depressive symptoms in the short term (moderate-quality evidence). A complement to treatment, not a replacement. (Aalbers et al. 2017; Tang et al. 2020)

Music therapy in clinical settings. Credentialed music therapy shows real but often modest benefits for specific outcomes in cancer care, dementia and autism. These results do not transfer to sound baths. (Bradt et al. 2021; van der Steen et al. 2025; Geretsegger et al. 2022)

Preliminary — Early or mixed evidence

Blood pressure & heart rate variability. Music can briefly lower heart rate and blood pressure. Reviews could not show it treats high blood pressure, and HRV studies carry a high risk of bias. (Bradt et al. 2013; Kühlmann et al. 2016; Mojtabavi et al. 2020)

Singing bowls. Small studies suggest people feel calmer after a session, but two systematic reviews found the evidence too thin or too biased to recommend bowls as therapy. (Stanhope et al. 2020; Cai et al. 2025)

Gongs & sound baths. We found no controlled gong trials. Two uncontrolled reports from one research group found people felt less tense and in a better mood after sound-meditation sessions. (Goldsby et al. 2017; Goldsby et al. 2022)

Binaural beats. A medium effect in lab studies of memory, attention, anxiety and pain, but more EEG studies contradicted "brainwave entrainment" than supported it. (Garcia-Argibay et al. 2019; Ingendoh et al. 2023)

Vibroacoustic therapy. A scoping review calls the research "too sparse" to say what helps pain. A 2019 fibromyalgia trial was retracted in 2020 for lacking a sham control. (Kantor et al. 2022)

40 Hz sound & light (gamma stimulation). Promising in mice; safe and well tolerated in small human pilot studies. Not shown to prevent or treat dementia in people. (Martorell et al. 2019; Chan et al. 2022)

Not supported — Not supported

432 Hz & solfeggio frequencies. Only tiny pilots (9–33 people) with small, short-term differences. Nothing supports DNA repair or "healing frequency" claims. (Calamassi et al. 2019; Akimoto et al. 2018)

Chakras, energy fields & "frequency healing". Relies on an energy field that has never been demonstrated; in a blinded test, practitioners did no better than chance. (NCCIH 2018; Rosa et al. 1998)

What the research does *not* show

- That any sound, instrument or frequency treats, cures or prevents a disease.
- That music-therapy results carry over to sound baths, bowls or recordings. Music therapy is delivered by a credentialed therapist.
- That a specific frequency (432 Hz, 528 Hz, 40 Hz) does something special in people, or that sound "balances" energy or chakras.
- That benefits last. Most effects were measured during or right after a session.

How to talk about it

SAY	"Many people find sound sessions relaxing, and music has good evidence for easing stress and anxiety in the moment."
AVOID	"Sound healing cures anxiety," "this frequency repairs DNA," or "a gong bath resets your nervous system."
SAY	"Music can be a helpful add-on alongside your usual care."
AVOID	Suggesting anyone replace medical treatment, medication or therapy with sound.

Limitations

Most music trials can't blind participants, so expectation explains part of many effects, and reviews often rate them at high risk of bias. Effects are usually small to medium and short-term, and much of the evidence is self-reported. Ratings are our editorial judgment of published reviews and trials, summarised mainly from abstracts; one cited trial has since been retracted and is not counted. Research changes, so check the dated sources online.

Sources: all 39 are listed with DOI and PubMed links at sound-evidence-hub.netlify.app/evidence/. Links, metadata and retraction notices were checked automatically on October 7, 2026: 0 dead links, 1 retraction (labelled), 0 metadata mismatches.

A companion resource to The Meta Musician. Educational only. It does not diagnose, treat, cure or prevent any disease.